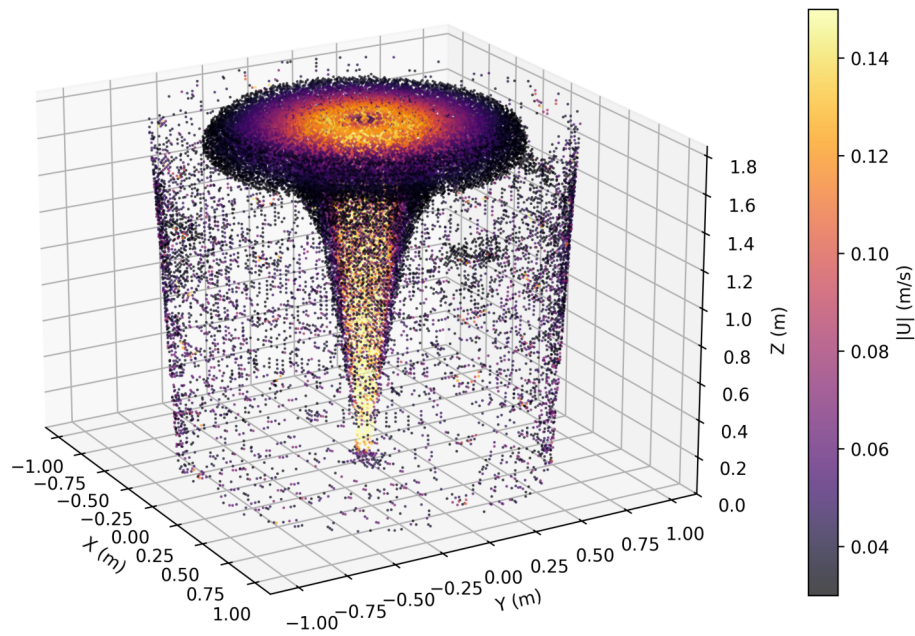


Center-bottom blowing: plume core ($|U| > 0.03$ m/s, steel phase)

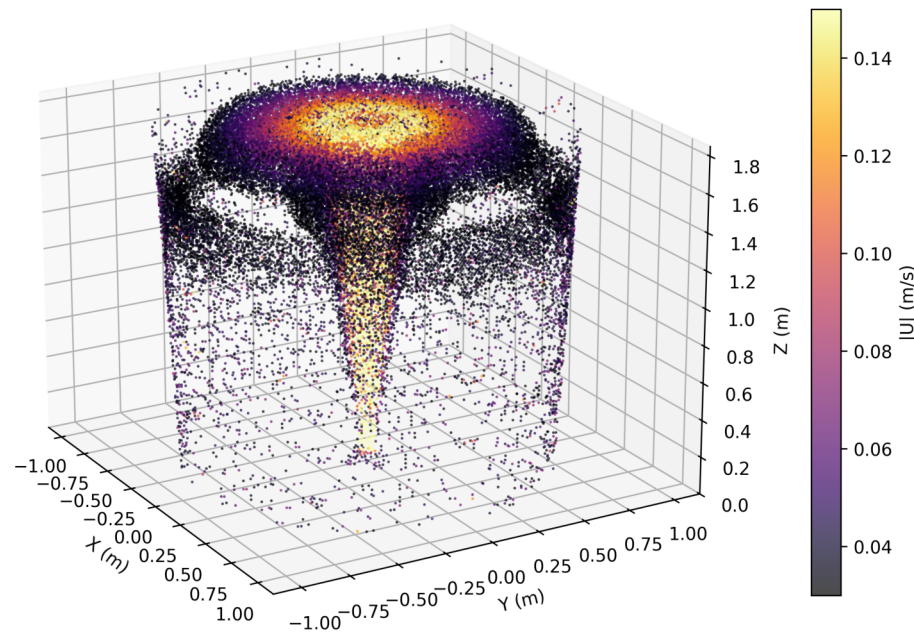
(a) Q=40 NL/min (Train)

Q=40 NL/min | Plume core ($|U| > 0.03$ m/s)



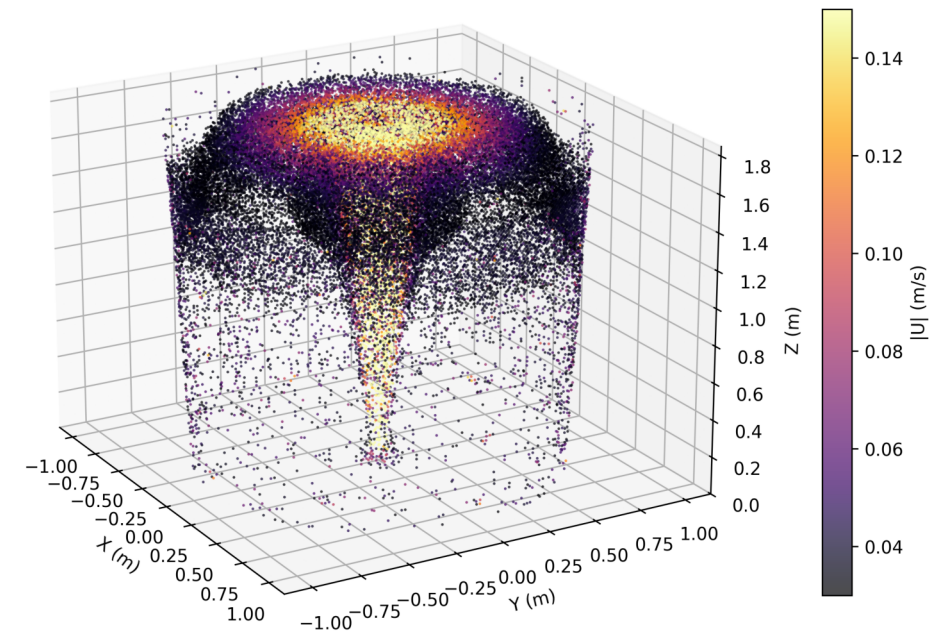
(b) Q=60 NL/min (Train)

Q=60 NL/min | Plume core ($|U| > 0.03$ m/s)



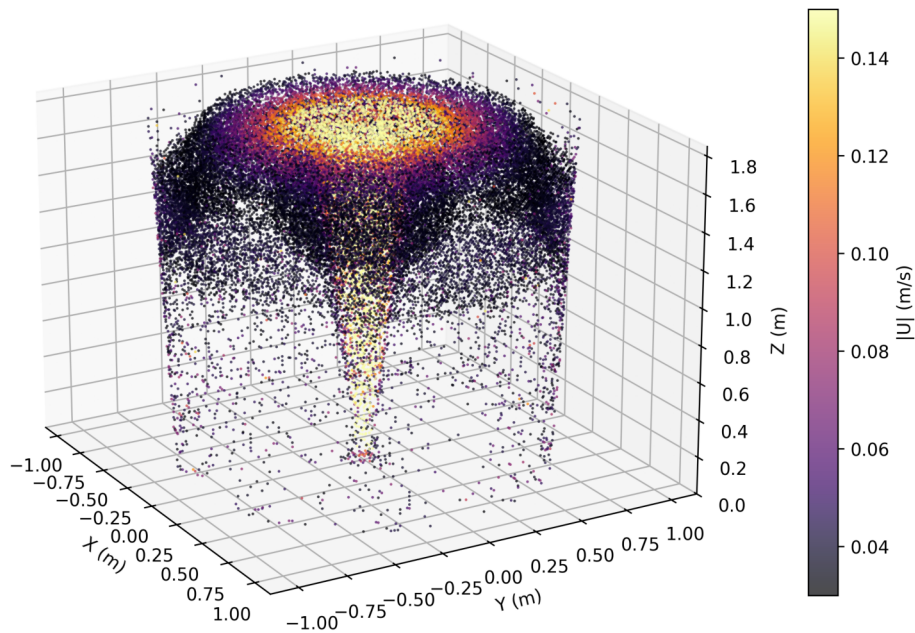
(c) Q=80 NL/min (Train)

Q=80 NL/min | Plume core ($|U| > 0.03$ m/s)



(d) Q=100 NL/min (Extrap.)

Q=100 NL/min | Plume core ($|U| > 0.03$ m/s)



(e) Q=120 NL/min (Extrap.)

Q=120 NL/min | Plume core ($|U| > 0.03$ m/s)

